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EXERCISE

3BCIUB

I. Multi Choice Question

1. The center of mass of a system of particles does not depend upon,

(a) position of particles

(b) relative distance between particles

(c) masses of particles

(d) force acting on particle

[AIPMT 1997, AIEEE 2004]

2. A couple produces,

(a) pure rotation

(b) pure translation

(c) rotation and translation

(d) no motion

[AIPMT 1997]

3. A particle is moving with a constant velocity along a line parallel to positive X-axis. The magnitude of its angular momentum with respect to the origin is,

(a) zero

(b) increasing with x

(c) decreasing with x

(d) remaining constant

[IIT 2002]

4. A rope is wound around a hollow cylinder of mass 3 kg and radius 40 cm. What is the angular acceleration of the cylinder if the rope is pulled with a force 30 N?

(a) 0.25 rad s^{-2} (b) 25 rad s^{-2}

(c) 5 m s^{-2} (d) 25 m s^{-2}.

[NEET 2017]

5. A closed cylindrical container is partially filled with water. As the container rotates in a horizontal plane about a perpendicular bisector, its moment of inertia,

(a) increases (b) decreases

(c) remains constant

(d) depends on direction of rotation.

[IIT 1998]

6. A rigid body rotates with an angular momentum L. If its kinetic energy is halved, the angular momentum becomes,

(a) L (b) L/2

(c) **2L** (d) $L/\sqrt{2}$

[AFMC 1998, AIPMT 2015]

7. A particle undergoes uniform circular motion. The angular momentum of the particle remain conserved about,

(a) the center point of the circle.

(b) the point on the circumference of the circle.

(c) any point inside the circle.

(d) any point outside the circle.

[IIT 2003]

8. When a mass is rotating in a plane about a fixed point, its angular momentum is directed along,

(a) a line perpendicular to the plane of rotation

(b) the line making an angle of 45° to the plane of rotation

(c) the radius

(d) tangent to the path

[AIPMT 2012]

9. Two discs of same moment of inertia rotating about their regular axis passing through center and perpendicular to